

Mechanical interference measures phase without fixing an action unit

An ideal classical string network sends a finite pulse down two paths and transfers its entire energy into two mechanical absorbers. Their work records contain an interference cross term. Attenuating the incident displacement leaves the normalized fringe unchanged while every energy and canonical action shrinks quadratically. This exploratory Q08 construction tests the physical phase premise left by Q07; it is not promoted to the claim ledger.

1. Wave, junction and receiver premises

Take identical linear strings with mass per length $\mu > 0$, tension $T > 0$, transverse displacement $u(x, t)$ and small slopes. Position and displacement have length units. Define speed $c = \sqrt{T/\mu}$ and mechanical impedance $Z = \sqrt{\mu T} = \mu c$, with units mass/time. The stipulated continuum model is

$$L = \int \frac{1}{2}(\mu u_t^2 - T u_x^2) dx, \quad \mu u_{tt} = T u_{xx}, \quad e = \frac{1}{2}(\mu u_t^2 + T u_x^2), \quad j = -T u_t u_x.$$

Multiplying the equation by u_t gives $\partial_t e + \partial_x j = 0$. A right-moving displacement $f(t - x/c)$ has $e = \mu(f')^2$ and $j = Z(f')^2$. Prepare a nonzero smooth compactly supported pulse f ; its incident energy is $E = Z \int_{\mathbb{R}} (f')^2 dt > 0$. This preparation supplies energy; no source operates during its subsequent passage.

A lossless four-port junction mixes two input strings into two output strings with the real matrix

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad S = \begin{pmatrix} 0 & H \\ H & 0 \end{pmatrix}.$$

Here S maps incoming to outgoing displacement profiles on four equal-impedance ports. It is symmetric, $S^2 = I$, and orthogonal. This is an ideal mechanical constraint, not an assertion that an ordinary tied-string knot has this rule. For an explicit boundary realization orient all four port coordinates outward, write $u = a(t + x/c) + b(t - x/c)$ and put $P_{\pm} = (I \pm S)/2$. Impose $P_- u(0, t) = 0$ and $P_+ u_x(0, t) = 0$. The first fixes the constrained junction displacement; the second supplies free force balance in its orthogonal subspace. For pulses initially at rest these equations give $b = Sa$. Indeed they give $P_- b = -P_- a$ and $P_+ b' = P_+ a'$; the integration constant vanishes before arrival. The displacement velocity lies in the plus subspace and the boundary force in the minus subspace, so their scalar product and boundary work vanish. Equivalently $\sum_j (a'_j)^2 = \sum_j (b'_j)^2$. An ideal massless linkage enforcing these fixed linear constraints is a model premise. Finite junction inertia and bandwidth are outside this construction.

At each final output install a viscous receiver with force coefficient Z . At a right endpoint the balance $T u_x = -Z u_t$ admits a pure outgoing wave without reflection. Work delivered to the receiver is

$$Q(t) = \int_{-\infty}^t Z u_t(s)^2 ds, \quad \dot{Q} \geq 0.$$

The dashpot represents a sink with an internal energy record Q ; field energy plus these records is conserved after preparation. It is not a finite closed Hamiltonian model of the bath. A semi-infinite matched string supplies an alternative conservative energy sink, with Q the energy transported into it. Neither realization imposes a minimum resolvable work or a discrete event rule.

2. Exact two-path energy readout

Use two such junctions. The first splits the input $(f, 0)$; the two arms have positive travel times τ_1, τ_2 ; the second mixes their arriving waves. The unused source port has no incoming wave and is matched. All connections are reflectionless in the stipulated model. Output displacements are

$$b_{\pm}(t) = \frac{1}{2}[f(t - \tau_1) \pm f(t - \tau_2)].$$

Integrating the receiver work over the complete pulse gives

$$Q_{\pm} = \frac{E}{2} \pm \frac{Z}{2} \int f'(t - \tau_1) f'(t - \tau_2) dt, \quad Q_+ + Q_- = E. \quad (1)$$

For $\delta = \tau_2 - \tau_1$, define the normalized derivative autocorrelation

$$C(\delta) = \frac{\int f'(s) f'(s - \delta) ds}{\int (f')^2 ds}, \quad \frac{Q_{\pm}}{E} = \frac{1 \pm C(\delta)}{2}. \quad (2)$$

Cauchy-Schwarz gives $|C| \leq 1$, so both receiver energies are nonnegative. At equal delays $C = 1$ and all energy reaches the plus output. When the two derivative supports do not overlap $C = 0$ and each receiver takes $E/2$. These are deterministic energies in a single pulse experiment. Dividing by E does not turn them into probabilities of exclusive detector clicks.

For the coherent limit choose $f_D(t) = Ag(t/D) \cos(\omega t)$ with a fixed nonzero smooth compactly supported envelope g , $\omega > 0$, and fixed delay δ . Let $D \rightarrow \infty$ at fixed A, ω, δ . The leading derivative is $-A\omega g(t/D) \sin(\omega t)$; the envelope derivative contributes lower order terms to the integrals divided by D . Setting $t = Dr$, translation of the envelope tends to the identity, while integration by parts makes the oscillatory double-frequency term vanish. Thus

$$C_D(\delta) \longrightarrow \cos(\omega\delta), \quad Q_+/E \longrightarrow \cos^2(\omega\delta/2), \quad Q_-/E \longrightarrow \sin^2(\omega\delta/2). \quad (3)$$

This is a long coherent pulse limit, with energy growing with D at fixed amplitude, not an exact sinusoidal formula for an arbitrary finite pulse. Equations (1)–(2) already give the finite-energy result. Scaling A with D to hold the energy fixed leaves the normalized correlation unchanged.

3. Which mechanical action could the phase measure?

A carrier delay produces the unwrapped relative phase $\phi = \omega\delta$; the two energy records measure its cosine. They determine neither its sign nor its winding without additional phase scans or timing information. Complex phasors may abbreviate the real sine/cosine solution; no quantum probability postulate enters (1).

To expose the action normalization, take the right-moving monochromatic wave $u = A \cos(\omega t - kx)$, $k = \omega/c$, and one wavelength $\ell = 2\pi c/\omega$. This is a periodic carrier calculation, also applicable to an exact carrier plateau of a sufficiently long pulse while the cell stays inside that plateau. With momentum density $p = \mu u_t$, its cell energy is

$$E_{\ell} = \int_0^{\ell} e dx = \frac{1}{2} \mu A^2 \omega^2 \ell = \pi Z A^2 \omega.$$

Over one temporal cycle the canonical loop integral and action variable are

$$J_{\ell} = \int_0^{2\pi/\omega} dt \int_0^{\ell} p u_t dx = \frac{2\pi E_{\ell}}{\omega}, \quad I_{\ell} = \frac{J_{\ell}}{2\pi} = \frac{E_{\ell}}{\omega} = \pi Z A^2. \quad (4)$$

Spatial integration over a complete wavelength makes $\int p u_t dx = E_{\ell}$ constant at every time. The signed canonical accumulation over a delay is $W_{\ell}(\delta) = E_{\ell} \delta$, hence

$$\phi = \frac{W_\ell(\delta)}{I_\ell}. \quad (5)$$

This supplies an explicit action-to-phase conversion, with a preparation- and impedance-dependent coefficient. Choosing a cell of n wavelengths multiplies both W and I by n without altering the phase. Moreover the spacetime Lagrangian action of this pure traveling wave is zero: $\mu u_t^2 = T u_x^2$ pointwise. Equation (5) uses a canonical integral, not $\int L dt$. A claim that the receiver measures $\exp(iS/K)$ must specify its action functional and boundary convention; equal action units do not justify identifying these quantities.

4. Decisive attenuation and apparatus tests

At fixed strings, junctions, delays and receivers replace f by ϵf with $0 < \epsilon \leq 1$. Linearity preserves all travel times and the normalized correlation exactly, including for finite pulses. Equations (1) and (4) give

$$E \mapsto \epsilon^2 E, \quad Q_\pm \mapsto \epsilon^2 Q_\pm, \quad I_\ell \mapsto \epsilon^2 I_\ell, \quad \phi \mapsto \phi. \quad (6)$$

Every positive amplitude retains a defined normalized fringe; its absolute work vanishes in the zero-amplitude limit. No measurable signal is asserted at the zero endpoint. A fixed receiver resolution would end practical visibility earlier and would supply an apparatus energy threshold.

A second family rescales μ, T, Z together by $\eta > 0$, including receiver impedances. The speed and displacement equations stay fixed, whereas energies and action integrals acquire η . A calibrated force or mass standard would distinguish this family. It demonstrates the same common normalization freedom as Q06 in an apparatus that now performs interference and absorbs work. Changing frequency at fixed delay changes the phase; at fixed cell energy it also changes $I_\ell = E_\ell/\omega$. Neither a cosine fringe nor a square-law energy record fixes that ratio universally.

5. Consequence and next decision

Within this model, coherent path composition and mechanical energy readout are compatible with arbitrarily small positive classical action. This closes Q07's proposed linear-wave receiver test. The result is a physical countermodel to selecting an action unit from interference alone, not a reconstruction of quantum detection. The admitted field has infinitely many modes and its receivers record divisible energy.

Park further linear splitter and impedance variants. The sharper next test is whether localized event readout excludes this alternative: construct a passive receiver with stored energy and a threshold, account for its reset work, and test whether exclusive events and their energy scale survive changes in receiver preparation. A threshold could sharpen event formation while still supplying its own scale; the calculation must decide that distinction. Compare this with the gap track before dispatch: the string model already specifies real physical dynamics, but its nondispersive relation $\omega = c|k|$ admits low frequencies as wavelength grows and supplies no positive infinite-volume frequency gap. Another linear mode calculation would not resolve the missing physical quantum Hamiltonian of G04.

Written consistency checks: variation and local energy balance in section 1; zero junction work by orthogonal boundary subspaces; receiver work and the sum/difference cancellation in (1); Cauchy-Schwarz and the envelope limit; spatial and temporal integrations in (4); distinct canonical and Lagrangian actions; exact attenuation and common-parameter rescaling. No computational numerical or symbolic verification was used.

Source context is the B18 companion and checkerboard comparison: their norm/probability construction motivates the comparison but does not supply the string junction or receiver model.

Q07 isolates the phase premise; Q06 isolates reciprocal calibration. This note gives its own elementary derivation and coordinator consistency review. Independent proof review and bounded literature comparison remain required before ledger promotion; no novelty claim is made.